



Quadrant Selection

CPU TIME LIMIT

1 second

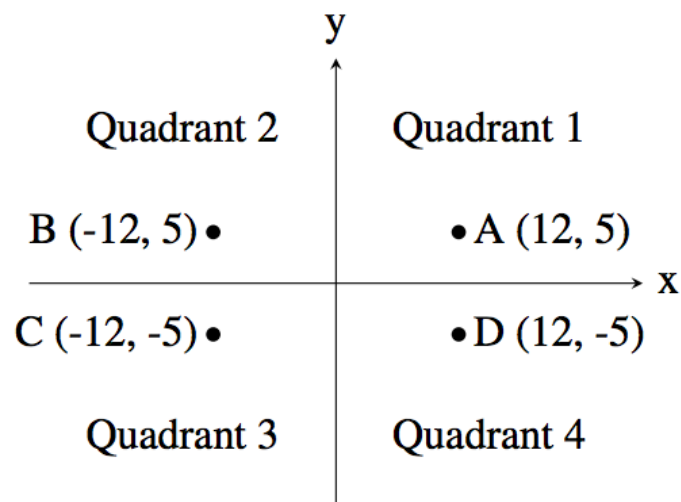
MEMORY LIMIT

1024 MB

DIFFICULTY

easy (1.2)

A common problem in mathematics is to determine which quadrant a given point lies in. There are four quadrants, numbered from 1 to 4, as shown in the diagram below:



For example, the point A , which is at coordinates $(12, 5)$ lies in quadrant 1 since both its x and y values are positive, and point B lies in quadrant 2 since its x value is negative and its y value is positive.

Your job is to take a point and determine the quadrant it is in. You can assume that neither of the two coordinates will be 0.

Input

The first line of input contains the integer x ($-1000 \leq x \leq 1000$; $x \neq 0$). The second line of input contains the integer y ($-1000 \leq y \leq 1000$; $y \neq 0$).

Output

Output the quadrant number (1, 2, 3 or 4) for the point (x, y) .

Sample Input 1

```
10
6
```

Sample Output 1

```
1
```

Sample Input 2

```
9
-13
```

Sample Output 2

```
4
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